

Satwik Medipalli

Data Scientist | Analytics, Forecasting & Applied ML

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EXPERIENCE

AI Researcher (UTD Client Project)

Jan 2026 – May 2026

DFW Technology

Richardson, TX

- Completed a semester-long UTD educational engagement on-site at DFW Technology, authoring *Public vs Private AI: Scalable Business Agents for SMEs* on Private AI-first architecture, governance, and compliance.
- Authored *When to Use Small Language Models: A Strategic Playbook*, including a Unit of Work pilot measuring >90% inference-cost reduction vs LLM-only at matched task quality.

AI Research Intern

Jan 2024 – Apr 2024

Boltzmann Labs

Hyderabad, India

- Built a multi-omic analysis pipeline over millions of records using clustering, differential expression analysis, and GSEA to prioritize cancer-linked cell populations for downstream validation.
- Containerized and deployed the pipeline with Docker, FastAPI, and Kubernetes, cutting researcher setup from hours to minutes.

Data Analytics Intern

Aug 2023 – Dec 2023

Bostonlogix

Hyderabad, India

- Migrated 1M+ Hungry Jack's records from MongoDB to SQL Server with Python, also built Bold BI dashboards so the client could monitor KPIs on the structured data.
- Benchmarked Azure OpenAI against classical NLP for recipe extraction on 100K Hungry Jack's records using precision and recall; the team adopted the LLM path in production, replacing manual ingredient review.

BI Developer Intern

Apr 2023 – Jul 2023

Ideabytes

Hyderabad, India

- Built live Power BI dashboards from IoT JSON APIs for the CEO and the company website, replacing manual weekly reporting.
- Forecasted one-year performance across 100+ IoT sensors with ARIMA in Python, shifting maintenance from reactive to scheduled.

PROJECTS

SiliconSentinel | [Live Demo](#)

2025–2026

Python, XGBoost, lifelines, SHAP, DuckDB, Next.js

- Built a fleet reliability stack on a physics-grounded simulator (Arrhenius / Black / Peck / Coffin-Manson): survival models, XGBoost 30-day failure risk, RUL regression, and SHAP explanations for ops triage.
- 30-day failure classifier hits ~83% recall at 80% precision, catching failures early (holdout PR-AUC 0.89; walk-forward 0.93 $\pm$ 0.04); Cox concordance 0.74 on the simulated fleet.

ShapeShift | [Live Demo](#)

2025–2026

Python, DuckDB, scikit-learn, Statcast, Next.js

- Built a pitch-design decision system over 5.6M+ Statcast pitches: outcome-blind change detection, then a locked matched event study so Keep / Reshape / De-emphasize calls rest on out-of-sample value, not Stuff+ vibes.
- Locked 2025 aggregate ATE +0.02 RV/100 (near null); Muñoz slider case at matched -0.29 RV/100 shows when reshape hurts.

NFL Draft Intelligence | [Live Demo](#)

2025

XGBoost, SHAP, lifelines, Next.js

- Built draft prospect models on 2000–2025 Combine and draft history with position-specific XGBoost classifiers and SHAP so scouts see *why* a projection moves.
- Added Kaplan-Meier / survival views for career longevity so evaluation covers durability, not only day-one grade.

Lumen | [Live Demo](#)

Summer 2025

React, TypeScript, FastAPI, D3, Yahoo, SEC EDGAR, FRED

- Shipped an S&P 500 research terminal over 503 names with live quotes, SEC XBRL (12 ratios), FRED macro, and risk tools (screener, correlation, backtest vs SPY).
- Mapped 955 relationship edges across the index; AI only narrates backend-computed evidence, and a Sources page refuses invented numbers when a feed is missing or stale.

EDUCATION

University of Texas at Dallas

Aug 2024 – May 2026

M.S. Business Analytics

Richardson, TX

GITAM University

Aug 2020 – May 2024

B.Tech Computer Science with Data Science

Hyderabad, India

TECHNICAL SKILLS

ML & Stats: XGBoost, Scikit-learn, SHAP, survival analysis (lifelines), ARIMA, clustering, GSEA

Data & Analytics: Python, SQL, Pandas, NumPy, DuckDB, MongoDB → SQL Server migrations, Power BI, Bold BI

Experimentation: matched event studies, precision/recall, PR-AUC, walk-forward validation

Software: FastAPI, Next.js, Docker, Kubernetes, Git, Azure